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$$\text{Mass of glucose} = \frac{\text{Molecular mass} \times \text{number of particles}}{\text{Avogadro's number}}$$

$$= (180 \times 5 \times 10^{23}) / 6.023 \times 10^{23}$$

$$= 149.43 \text{ g}$$

IV. Calculation based on number of atoms/ molecules.

1) Calculate the number of molecules in 11.2 litre of CO_2 at S.T.P

$$\text{Number of moles of } CO_2 = \frac{\text{Volume at S.T.P}}{\text{Molar volume}}$$

$$= 11.2 / 22.4$$

$$= 0.5 \text{ mole}$$

Number of molecules of CO_2 = number of moles of CO_2 × Avogadro's number

$$= 0.5 \times 6.023 \times 10^{23}$$

$$= 3.011 \times 10^{23} \text{ molecules of } CO_2$$

2) Calculate the number of atoms present in 1 gram of gold (Atomic mass of Au = 198)

$$\text{Number of atoms of Au} = \frac{\text{Mass of Au} \times \text{Avogadro's number}}{\text{Atomic mass of Au}}$$

$$\text{Number of atoms of Au} = \frac{1}{198} \times 6.023 \times 10^{23}$$

$$\text{Number of atoms of Au} = 3.042 \times 10^{21} \text{ g}$$

3) Calculate the number of molecules in 54 gm of H_2O?

$$\text{Number of molecules} = \frac{(\text{Avogadro number} \times \text{Given mass})}{\text{Gram molecular mass}}$$

Number of molecules of water $= 6.023 \times 10^{23} \times 54 \ / \ 18$

$$= 18.069 \times 10^{23} \text{ molecules}$$

4) Calculate the number of atoms of oxygen and carbon in 5 moles of CO_2.

- 1 mole of CO_2 contains 2 moles of oxygen
- 5 moles of CO_2 contain 10 moles of oxygen

Number of atoms of oxygen = Number of moles of oxygen × Avogadro's number

$$= 10 \times 6.023 \times 10^{23}$$

$$= 6.023 \times 10^{24} \text{ atoms of Oxygen}$$

- 1 mole of CO_2 contains 1 mole of carbon
- 5 moles of CO_2 contains 5 moles of carbon

$$\text{No. of atoms of carbon} = \text{No.of moles of carbon} \times \text{Avogadro's number}$$

$$= 5 \times 6.023 \times 10^{23}$$

$$= 3.011 \times 10^{24} \text{ atoms of Carbon}$$

V. Calculation based on molar volume

Calculate the volume occupied by:

1) 2.5 mole of CO_2 at S.T.P

$$\text{Number of moles of } CO_2 = \frac{\text{Given volume at S.T.P}}{\text{Molar volume at S.T.P}}$$

$$2.5 \text{ mole of } CO_2 = \frac{\text{Volume of } CO_2 \text{ at S.T.P}}{22.4}$$

Volume of CO_2 at S.T.P = 22.4×2.5

$$= 56 \text{ litres.}$$

2) 12.046×10^{23} of ammonia gas molecules

$$\text{Number of moles} = \frac{\text{Number of molecules}}{\text{Avogadro's number}}$$

$$= 12.046 \times 10^{23} / 6.023 \times 10^{23}$$

$$= 2 \text{ moles}$$

Volume occupied by NH_3

= number of moles × molar volume

$= 2 \times 22.4$

$= 44.8$ litres at S.T.P

3) 14 g nitrogen gas

Number of moles = 14 / 28

$= 0.5$ mole

Volume occupied by N_2 at S.T.P

= no. of moles × molar volume

$= 0.5 \times 22.4$

$= 11.2$ litres.

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